

Oil Spill Detection & Vessel Attribution

Decision record — what was chosen, what was rejected, and what changed. Generated 26 August 2026.

0. In plain words — read this first

Ships sometimes dump oily waste at sea and sail away. Nobody is watching most of the ocean, so it usually goes unnoticed. This system watches it from a satellite.

How it works, in four sentences. A radar satellite photographs the sea. Oil flattens the water, so it shows up as a dark patch. Many other things also look dark — calm water, algae, rain — so we use the wind, the shape and the texture to work out which patches are really oil. Then, because oil drifts, we run the sea currents backwards to guess where it started, and list which ships were near that spot at that time.

The most important thing to understand. We produce a *ranked list of ships that could be responsible*. That is a lead for an investigator to follow, not proof. A ship being near a spill is a coincidence until somebody checks. If the evidence is weak, the system says "not enough evidence" instead of guessing — and today it says that 97% of the time, which is it working properly, not failing.

Jargon you will hear. SAR is the radar camera on the satellite. *Look-alike* is something dark that is not oil. AIS is the radio signal ships broadcast saying where they are. *Drift* is oil moving with the current after it spills. *Abstain* means we decline to name anyone.

1. What the system does, in more detail

A Sentinel-1 radar scene of the ocean goes in. Dark patches come out, each tested against physics to decide whether it is oil or a natural look-alike. Surviving slicks are drifted backwards through real ocean currents to estimate where they started, and AIS tracks near that origin are scored to produce a **ranked list of candidate vessels** — never a single accusation.

Deployed at oilspill.onrender.com: 16 scenes, 34 presented detections, 74 MB resident against a 512 MB limit.

2. Where the data comes from, and what we turned down

The binding constraint was access, not quality. Every source had to be obtainable without an institutional affiliation, and two datasets the original plan depended on turned out not to be.

Need	Chosen	Rejected, and why
SAR imagery	AWS Open Data mirror , catalogue via CDSE OData. Windowed <code>/vsicurl</code> reads: a scene costs 2 MB, not 1 GB.	CDSE download needs an account whose registration page would not render. Planetary Computer — Hub deprecated.
Wind	ERA5 via Open-Meteo . Same reanalysis, same 0.25° grid, open HTTP, no key.	Copernicus CDS — identical data, but blocks on registration and a <code>.cdsapirc</code> .
Currents	Copernicus Marine (CMEMS) , one NetCDF per scene.	No open alternative at this resolution. Needs a username and password; there is no API key, which is the usual confusion.
AIS positions	NOAA Marine Cadastre — the only free source here publishing real position reports.	GFW serves identity and events but <i>no tracks</i> . Danish DMA host unreachable. AISStream is live-only, so it cannot match a past scene. MarineTraffic / Spire are paid.
Dark vessels	GFW gap events , carrying the publisher's own <i>intentionalDisabling</i> assessment.	Nothing else publishes an adjudicated AIS switch-off. Coverage is thin: one gap in the whole Indian Ocean box over ten days.
Incidents	NOAA IncidentNews plus a curated world catalogue, 3,415 spills.	Used for corroboration only, never as training labels.

3. Which methods we used to find the oil

3.1 Finding the dark patches

The plan called for a U-Net with a ResNet-34 encoder, chosen because SkyTruth Cerulean uses it for this exact task — a complete answer to "why this architecture". It was trained, reached **0.70 oil IoU on balanced patches**, and then failed to transfer to whole scenes, where overlap with truth was near zero.

What runs instead: a classical adaptive dark-patch detector. The checkpoint is parked rather than deleted, and the running system reports *classical-dark-patch* on its own status panel rather than implying a deep model is active. A detector that works beats a better one that does not.

3.2 Deciding which dark patches are really oil

Deliberately **not** a neural network. A small logistic model over interpretable physical features means every rejection can be explained out loud: *"rejected: wind 1.2 m/s, below the 2 m/s floor"*. An explainable rejection is worth more here than a fractional IoU gain buried in a tensor.

Wind is kept **out** of the network for the same reason. Adding it as a fourth input channel would probably help slightly, and would make the wind-ablation claim unmeasurable.

4. The science rules we built in

4.1 Wind: a strict rule at one end, a flexible one at the other

The two ends of the detection window fail differently, and treating them identically was a bug. **Below** the floor, calm water is genuinely indistinguishable from oil — every dark patch is unexplainable, so a hard gate is right and it prevents false positives. **Above** the ceiling the failure inverts: oil becomes invisible, so a large, strongly damped, elongated feature seen anyway is *evidence*, and gating it converts a real detection into a miss.

The ceiling is now a graded penalty with a hard gate only at 15 m/s. This is what the brief asked for — "soft edges ... a graded feature, not a hard cutoff" — and the original code did not implement it.

4.2 The longer ago it spilled, the further away we look

A flat 60 km asked a slick found a fortnight after an incident to have stayed where it started, which is the one thing oil never does. Real MSC ELSA 3 detections 15 days later sat 87 km out and scored as uncorroborated. The radius is now 30 km plus 20 km per elapsed day, capped at 250 km — below the 0.3 m/s the project's own validation script quotes, so it errs toward matching too little rather than too much.

4.3 Never blame a ship for a leak that is already known about

The most serious bug found in this build. On the Taylor Energy MC-20 scene the system ranked three named vessels — FAST RUNNER, SM NEW ORLEANS, GULF DAWN — against oil from a wellhead that has leaked continuously since 2004. One did not even abstain. The brief calls this the worst failure the system can produce.

Two independent gaps allowed it. Morphology recognised a known fixed source only within 15 km, and the plume streams well past that. And the registry, which *had* identified the source correctly on every one of them, was never consulted for the decision, because `SpillIncident.to_dict` dropped the flag before it reached the pipeline. A documented fixed-source match now overrides the geometric test: an independent identification outranks a radius.

5. How the website runs on a free server

The free tier gives 512 MB. The pipeline needs far more, but the *result* is tiny — 730 KB of polygons and scores for 16 scenes, with no arrays at all. So the analysis runs once during docker build, where memory is plentiful, and the container only ever deserialises the answer.

Rejected: analysing per request (OOM-killed mid-request, which surfaces as a 502 with no body and no log line); a larger paid instance (unnecessary once the real memory bugs were found — one import alone was allocating 933 MB); a self-managed Oracle VM (trades a git push deploy for server administration the night before a demo).

The container also refuses live analysis outright. Without that, one "re-analyse" click loads the 933 MB coastline grid and kills the service.

6. What we changed our minds about, and why

Change	Why
Wind moved from a hardcoded constant to real ERA5	Both configs used a climatological constant sitting mid-window, so every candidate scored a perfect wind match and the strongest discriminator contributed nothing. 63 detections became 28 . Most of the loss was MSC ELSA 3, where real wind was 11.4–13.4 m/s: those detections existed only because the model was fed an invented 6.1 m/s.
ELSA 3 validated on a different scene	The 28 May pass is outside the wind window. A 9 June pass — same satellite, same orbit, 15 days after the sinking with the wreck still leaking, ERA5 wind 5.33 m/s — supports the claim honestly. All four validation checks now pass.
Currents moved to real CMEMS	With a synthetic field the drift-direction check was close to a coin flip, as the validation script said in its own output. Real currents produce about three times the correction, both toward the wreck.
Drift interpolation rewritten	xarray.interp rebuilt an intermediate Dataset per particle per timestep: 11.9 ms a sample, and one scene took 620 s against a two-minute target. Preloading into numpy made it 22 μ s — 4.6 s for the same scene, with a maximum disagreement of 2×10^{-9} m/s against the old path.
Unattributable detections split out of "active"	Of 26 current detections only one had a ranked vessel. Listing the other 25 alongside it implied leads that do not exist; they now sit under <i>Insufficient data</i> .
Synthetic scenes removed, then one restored and labelled	With 15 real scenes a fabricated one is a liability. But no real scene can demonstrate vessel ranking — no AIS, or a fixed source, or no slick — so exactly one returns, drawn dashed and labelled everywhere.
Abstention rate corrected from 100% to 97.1%	It counted the 211 rejected look-alikes as abstentions. A rejected candidate has no vessel by definition, so the denominator was swamped and the figure contradicted the map.

7. A thing we tried that did not work, and why we are saying so

The brief names look-alike false-positive rate as the primary decider, and the model ships hand-set weights. Fitting them on Yang et al. (PANGAEA 10.1594/PANGAEA.980773) would have turned a process claim into an accuracy claim. It was attempted and abandoned on evidence.

Three assumptions failed first. The **images are gated** — the bulk archive answers 401 and refers you to the principal investigator, exactly the situation the dataset had been chosen to avoid. There are **no k-means cluster labels**; the only subgroup split published is oil/no-oil × coast/water. What is open is the tabular metadata, and it names the source Sentinel-1 product for all 5,515 objects — so the extractor reads the *original calibrated imagery* off the AWS mirror instead of 8-bit JPGs, which is better than the dataset as distributed.

The blocker is annotation convention. Yang boxes oil objects but labels whole patches for look-alikes — reasonably, since there is nothing to box when the whole scene is the phenomenon. Median area is **1.1 km² for oil against 153.2 for look-alikes**, so log_area carries the strongest separation in the table while encoding nothing but annotation style. damping_db separates at **0.00**: a whole-patch label has no surrounding sea to compare against. A model fitted on that scores well on hold-out and is worse in production, so no model file was written.

One result survived. Wind is derived from position and timestamp, so the annotation asymmetry cannot touch it. Across 174 independently-labelled objects, look-alikes sit below the detection window **32% of the time against 21% for real oil** — roughly 1.5×. That is direct support for the premise the whole differentiator rests on, and it depends on no part of our own detector. Full write-up: docs/lookalike-fit-attempt.md.

8. What we claim, and what we refuse to claim

Defensible	Explicitly not claimed
Near-real-time detection, 3–24 h after acquisition	Real-time. Imagery lands 3–24 h late and free AIS lags ~72 h.
Probabilistic <i>ranked</i> attribution with an evidence bundle	Identifying the responsible vessel. Correlation is not evidence.
Dark-vessel flagging from adjudicated AIS gaps	Continuous monitoring. Revisit over one point is 6–12 days.
Every rejection explained in physical terms	Oil thickness, volume or type. SAR cannot measure them.
Validation against a real, documented Indian incident	A benchmark score. The look-alike model is not fitted, and the segmentation running is not the trained network.

9. Things this system cannot do

Written plainly, because these are the questions the system will be asked and it is better to answer them first. None of these are bugs. They are limits of radar, of free data, or of what a correlation can honestly support.

9.1 We cannot watch continuously

The satellite is not parked over India. It circles the Earth, and it passes over any given patch of sea roughly **once every 6 to 12 days**. Oil spreads, breaks up and disappears in days. So a ship could dump oil the morning after a pass, and by the next pass there would be nothing left to see. **We are taking occasional photographs, not watching a video**. Anyone who says otherwise about a free satellite service is mistaken.

9.2 We cannot see it as it happens

Even when the satellite does pass over, the picture takes **3 to 24 hours** to reach us: it has to be downlinked, processed and published. The free ship-tracking data is slower still, about **3 days** behind. So the honest description is "we find it soon after", never "we find it live". A spill we report this morning may have happened yesterday, and the ship may already be somewhere else.

9.3 We cannot see oil in the wrong wind

This is the one people find most surprising. Radar does not see oil directly — it sees that the sea is unusually *smooth* where the oil is. That only works in a middle range of wind.

If the wind is **too light** (under about 2 m/s), the whole sea is already smooth, so oil looks the same as everything around it. If the wind is **too strong** (above about 15 m/s), the waves churn the oil under and it stops flattening anything. Either way we see nothing.

The consequence matters: **"we found no oil" is not the same as "there was no oil"**. If the wind was wrong, we simply could not have seen it. The system records the wind with every result so this can always be checked.

9.4 We cannot tell how much oil there is

Radar tells us *something is there* and roughly what shape and area it covers. It cannot tell us **how thick the layer is**, so it cannot tell us the volume. A very thin sheen and a serious slick can look similar. It also cannot tell us **what kind** of oil it is — crude, diesel, fuel oil or vegetable oil all just look dark. Only a ship or an aircraft taking a sample can answer those.

9.5 We cannot prove who did it

This is the most important limit in the whole document. What the system actually finds is: *this oil probably started near here, around this time, and these ships were nearby*. That is a **coincidence in space and time**. It is a good reason for an investigator to look closer. It is not evidence.

Three things stop it from being proof. The backwards drift estimate has real uncertainty — typically tens of kilometres, and it grows the further back we go. Busy sea lanes have many ships passing the same point. And ship tracking can be switched off or falsified, so the guilty ship may simply not appear in the list at all. **We therefore publish a ranked list with confidence scores, and never name a single culprit**.

9.6 We often cannot name anyone at all

Right now **1 of our 34 detections** has any ships ranked against it. For the other 33 the system says "not enough evidence". That is deliberate. The reasons are: free ship data does not cover Indian waters in our sample; some slicks come from a known leaking wellhead where blaming a passing ship would be plainly wrong; and some had no ship anywhere near the estimated origin. **A system that always produces a name is a system that is guessing**.

9.7 We cannot see very small or very faint slicks

Below roughly 0.05 km² a dark patch is indistinguishable from ordinary radar noise, so it is discarded. Small deliberate discharges can slip under that floor. Raising the sensitivity would mean reporting noise as oil, which is worse.

9.8 We cannot work reliably close to shore

Near the coast, harbours, shallow water and land features all create dark and bright patterns that confuse the detector. Results in coastal water are weaker than in open sea, and our test data for coastal cases is thin — 19 samples out of 174.

9.9 What we have not finished

Two parts of the system are not what was planned, and the interface says so rather than hiding it. The AI segmentation model was trained and reached usable accuracy on small image tiles, but did not work on full scenes, so a simpler and more reliable method is running instead. And the oil-versus-look-alike scoring uses values set by hand from known physics rather than learned from data — we tried to learn them from a public dataset and stopped when we found that dataset could not support it honestly (section 7).

9.10 What radar CAN do, for balance

It sees through cloud, and it works at night. That is why radar is used for this at all: an ordinary camera would be blind half the time and blocked by weather the rest. The limits above are the price of that capability, not signs of a broken system.

The architecture diagram is a separate file: [docs/architecture.jpg](#).